

STATISTICAL BUSINESS ANALYSIS

Project Overview

A comprehensive statistical analysis of business sales data, including descriptive statistics, hypothesis testing, correlation analysis, confidence intervals, and regression analysis.

Objectives

- Calculate descriptive statistics for business metrics
- Test hypotheses about regional sales differences
- Analyze correlation between quantity and sales
- Calculate confidence intervals for key metrics
- Perform regression analysis to understand relationships

Setup & Installation

1. Install Python 3.x
2. Install required libraries:
`pip install pandas numpy scipy matplotlib seaborn`
3. Clone this repository
4. Open `statistical_analysis.ipynb` in Jupyter/VS Code
5. Run all cells in order

Project Structure

- `statistical_analysis.ipynb` — main analysis notebook
- `data/` — contains `business_data.csv`
- `visualizations/` — contains generated charts
- `hypothesis_tests_results.txt` — summary of all test results
- `statistical_report.pdf` — final report with insights

Technical Approach-

Descriptive Statistics

Calculated mean, median, mode, standard deviation, and variance for Total Sales.

Hypothesis Testing (3 Tests)

1. **T-Test** — Compared North vs South region sales
2. **ANOVA** — Compared sales across all regions
3. **Correlation Test** — Tested significance of Quantity-Sales relationship

Confidence Interval

Calculated 95% confidence interval for average sales using t-distribution.

Regression Analysis

Performed linear regression to model relationship between Quantity and Total Sales.

Key Findings

- Average Sales: ₹123,650.48 (95% CI: ₹103,776 - ₹143,524)
- No significant difference in sales across regions ($p = 0.0972$)
- Significant correlation between Quantity and Sales ($r = 0.69$)
- Regression shows Quantity explains ~47% of sales variation

Business Insights

1. Sales strategies don't need major regional customization
2. Focusing on quantity/volume can meaningfully drive sales
3. Other factors (beyond quantity) account for over half of sales variation — worth investigating (e.g., product type, pricing)

What I Learned

- How to perform and interpret hypothesis tests (t-test, ANOVA)
- Understanding p-values and statistical significance
- Calculating and interpreting confidence intervals
- Linear regression and R-squared interpretation
- Translating statistical results into business recommendations

Chart 1: Distribution of Total Sales

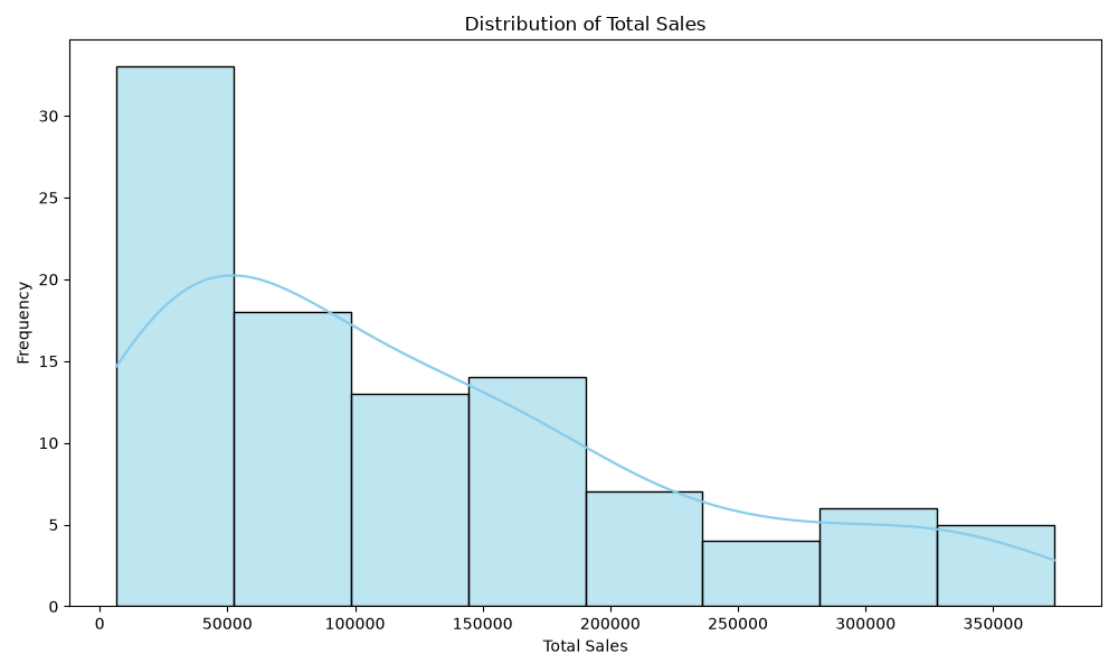


Chart 2: Correlation Heatmap

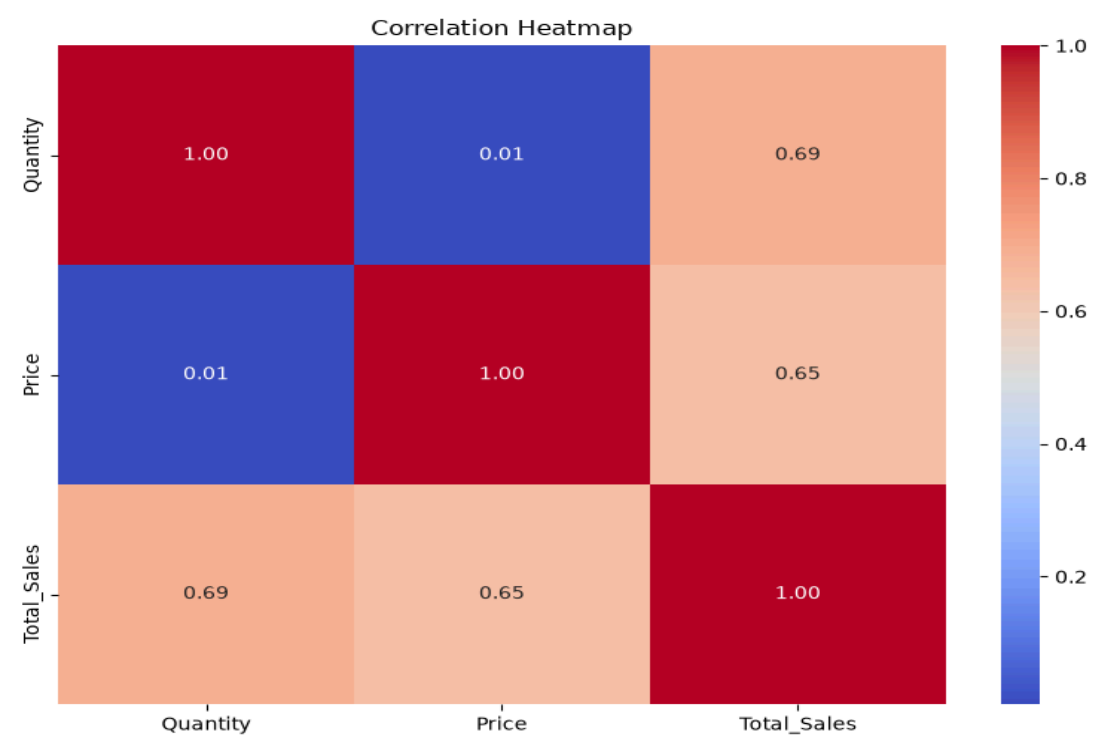


Chart 3: Regression Analysis

